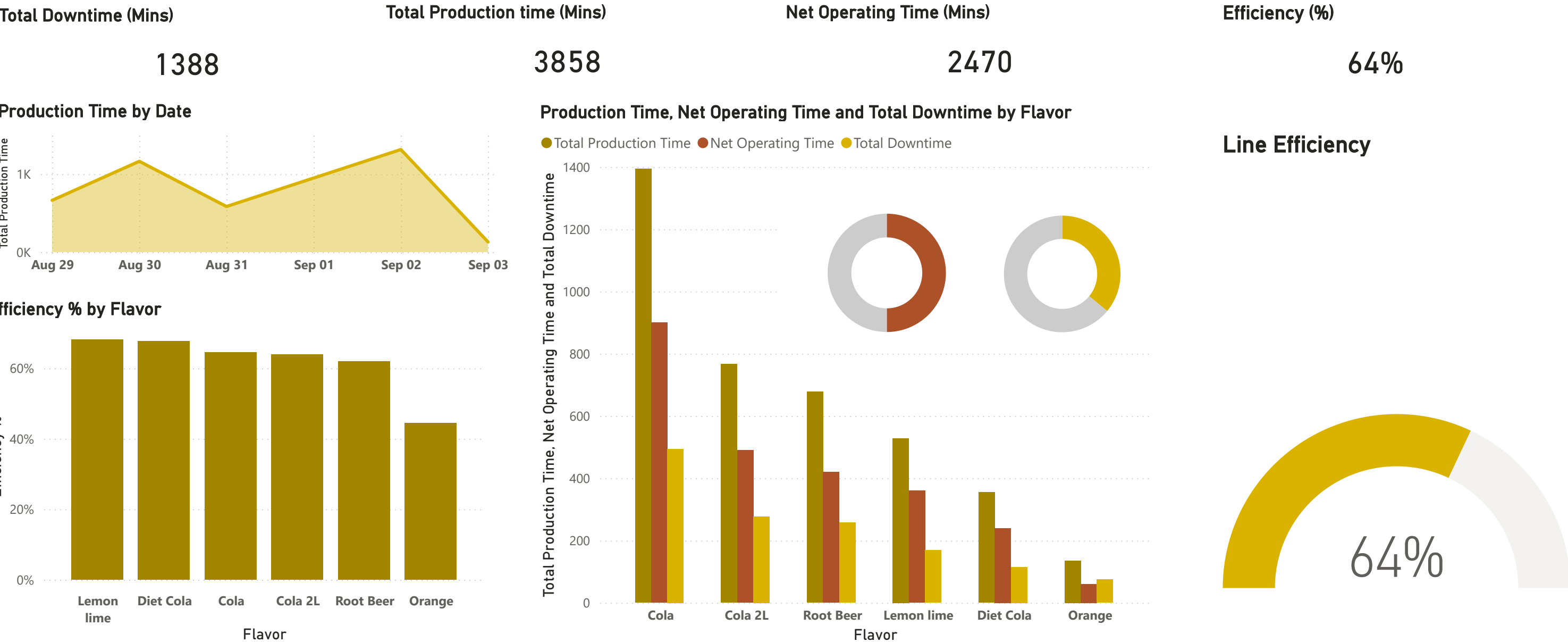


Manufacturing Downtime Analysis

Production Performance Overview:

This covers report on total production time, total downtime, and actual performance compared to expected standards.

Period Covered: 29 August - 3 September 2024



From **29 August to 3 September**, total production time was **3,858 minutes**, with **2,470 minutes** as net operating time and **1,388 minutes** lost to downtime, resulting in an **efficiency of 64%**.

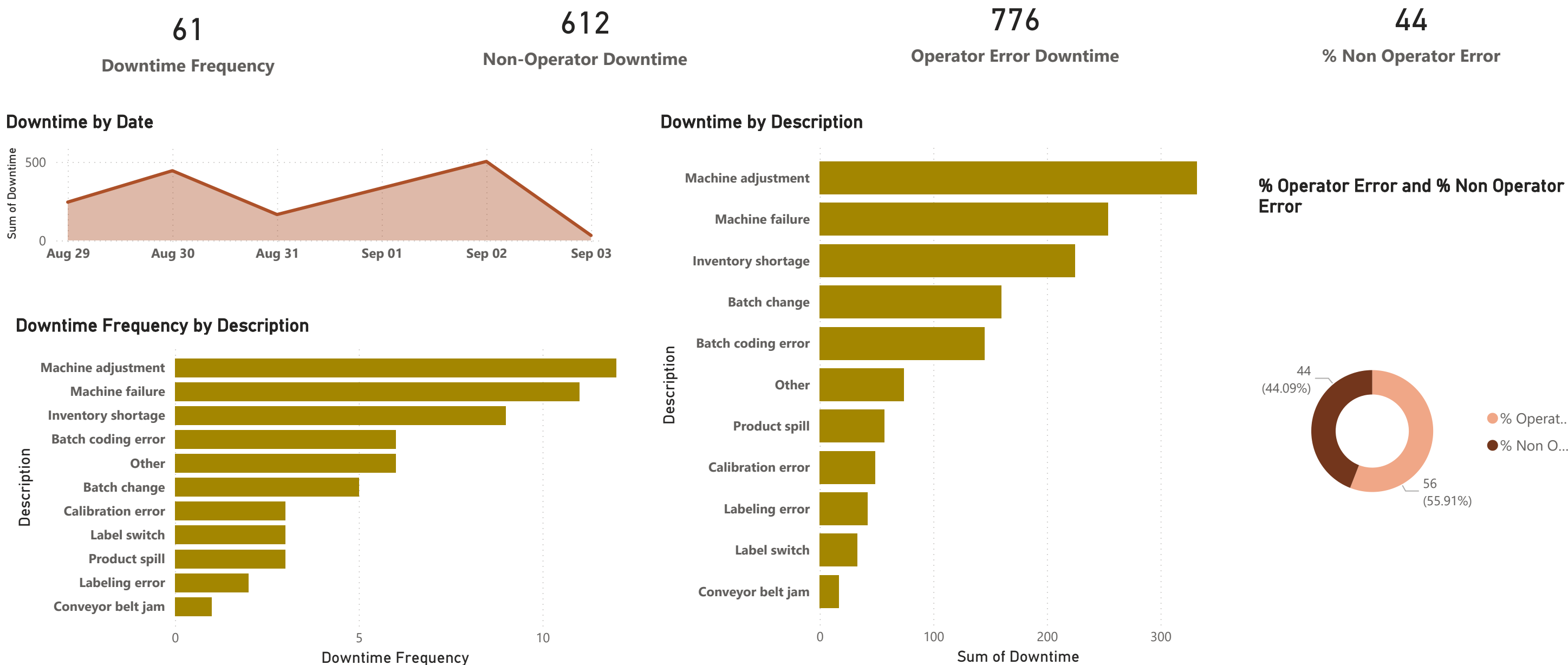
Production fluctuated over the period. **Lemon Line and Diet Cola recorded the highest efficiency (68%)**, followed by **Cola (65%)** with **Orange recording the lowest efficiency of 44%**. **Cola also had the highest production time and downtime**, followed by **Cola 2L**.

Overall, the analysis shows that **higher production time is associated with increased downtime**, indicating that high-volume lines may require better maintenance and optimization to improve efficiency.

Downtime Root Cause Analysis

What is slowing down production?
A breakdown of downtime causes, frequency, and whether issues are driven by operator errors or system inefficiencies.

Period: 29 August - 3 September 2024



A total of **61 downtime events** were recorded, with **operator-related downtime (776 mins) exceeding non-operator downtime (612 mins)**, accounting for **56% vs 44%** respectively. Downtime fluctuated across the period, indicating inconsistent operational performance.

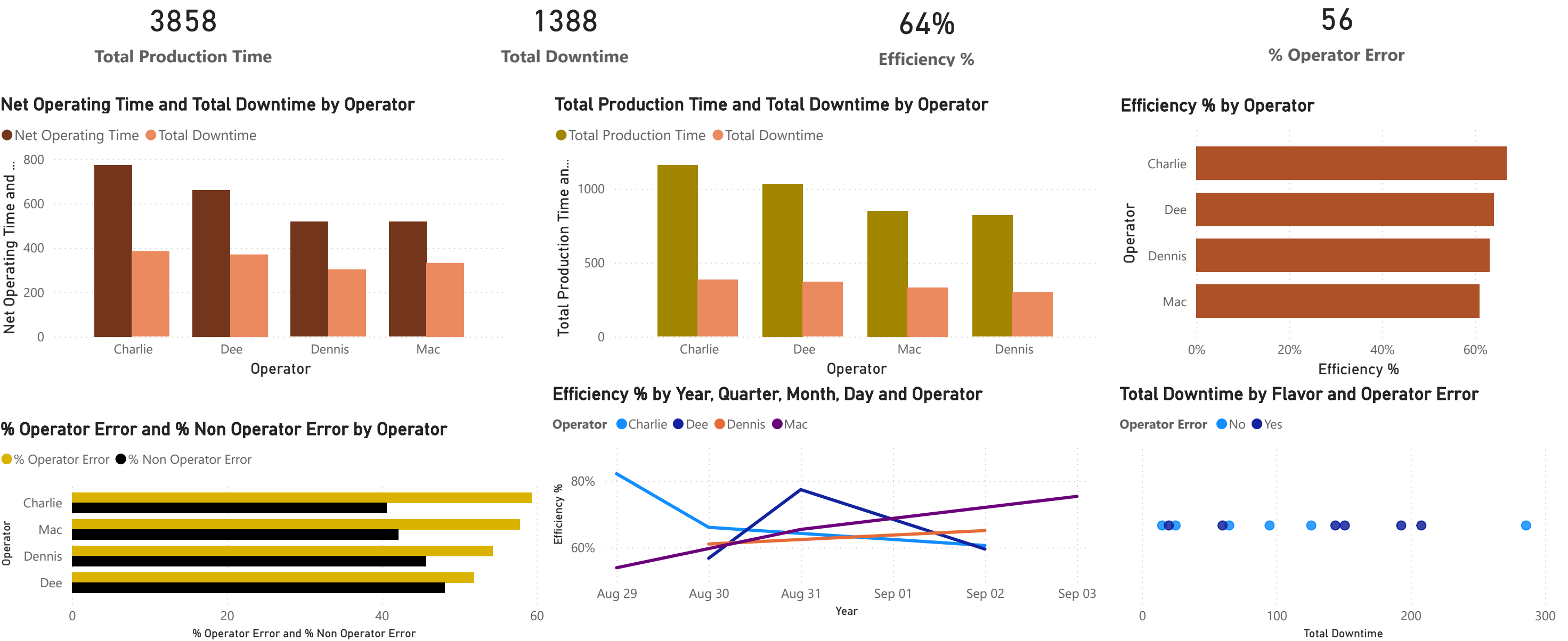
The main contributors to downtime were **machine adjustment (12 events, 332 mins)** and **machine failure (11 events, 254 mins)**, making them the most significant causes of production delays.

Overall, both human-related factors and machine issues are slowing down production, with operator errors contributing more to total downtime, while machine-related issues remain a major recurring cause.

Operator Deep Dive

Who is driving efficiency or delays?
Analyze individual operator performance, their impact on downtime, and the types of issues they encounter,

Period: 29 August - 3 September 2024



Charlie handled the highest production and operating time, as well as the most downtime, followed by **Dee, Dennis, and Marc**. Despite this, Charlie achieved the **highest overall efficiency (67%)**, while **Marc recorded the lowest (61%)**. Charlie also had the highest **operator-related errors**, whereas **Dee recorded the most non-operator errors**. At the start of the period, **Charlie led with an efficiency of 82%**, while **Marc had the lowest at 54%**. By the end, **Marc improved significantly to 75%**, while **Charlie’s efficiency declined to 61%**, with other operators maintaining relatively stable performance. Overall, efficiency varies by operator, with **Charlie driving high output but also higher errors**, and **Marc showing the most improvement over time**.